

OnRamps Precalculus Unit 4 Test Review

Part I – Fundamentals

Without this knowledge, you aren't passing the test!

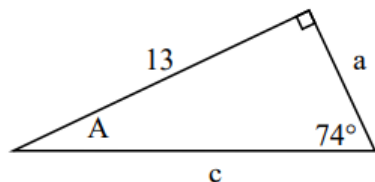
1. General Knowledge and Concept Questions

- a) What is the relationship between sine and cosecant, cosine and secant, and tangent and cotangent functions?
- b) What is the initial side and terminal side of an angle in standard position? What is a coterminal angle? What is a reference angle?
- c) How many degrees are there in a circle? How many radians? What is a radian (know the definition and relation to circumference of a circle)? How do you convert from degrees to radians and vice versa?
- d) What are the side-lengths of a general 30-60-90 and 45-45-90 right triangle?
- e) What is an identity? What is a formula? How are they different? List at least three identities you know. Give one or two examples of a formula.
- f) Memorize the unit circle and the Pythagorean identities.
- g) What is the domain and range of sine, cosine, and tangent functions? (Write in set notation!).

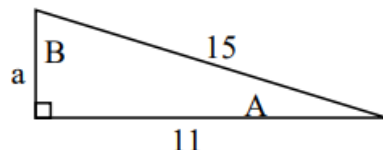
2. Simple Triangle Trigonometry

In each of the following triangles, solve for the unknown sides and angles.

a)



b)



3. Converting Between Degrees and Radians

Convert each angle measure to radians (if given in degrees) or degrees (if given in radians). Do not use a calculator to assist in simplifying your fractions.

- | | | | |
|---------------------|---------------------|----------------------|----------------------|
| a) 225° | b) 135° | c) 75° | d) 320° |
| e) $\frac{7\pi}{6}$ | f) $\frac{3\pi}{2}$ | g) $\frac{5\pi}{18}$ | h) $\frac{7\pi}{12}$ |

4. Coterminal Angles and Reference Angles

For each angle, give at least two coterminal angles (one positive, one negative) and identify the reference angle.

- | | | | |
|---------------------|----------------------|-----------------------|------------------------|
| a) $\frac{3\pi}{4}$ | b) $-\frac{5\pi}{6}$ | c) $\frac{29\pi}{12}$ | d) $-\frac{25\pi}{18}$ |
|---------------------|----------------------|-----------------------|------------------------|

5. Proving Basic Identities

Prove the other two Pythagorean identities using $\sin^2 \theta + \cos^2 \theta = 1$.

a) $\tan^2 \theta + 1 = \sec^2 \theta$

b) $\cot^2 \theta + 1 = \csc^2 \theta$

Prove the following using the Pythagorean identities and reciprocal identities.

b) $\frac{\tan \theta}{\sec \theta} = \sin \theta$

c) $\sin^2 \theta - \cos^2 \theta = 2 \sin^2 \theta - 1$

d) $\frac{\sin \theta \cdot \cot \theta}{\sec \theta} = \cos^2 \theta$

6. Find trigonometric ratios based on a given trigonometric value in a specified quadrant

a) For $\sin \theta = -\frac{\sqrt{5}}{3}$, and $\pi < \theta < \frac{3\pi}{2}$, find the other five trigonometric values.

b) For $\sec \theta = \frac{17}{8}$, and $\frac{3\pi}{2} < \theta < 2\pi$, find the other five trigonometric values.

7. Solving Basic Trig Equations

Solve each equation for $0 \leq \theta < 2\pi$.

a) $2 \cos \theta + 1 = 0$

b) $-2 \sin \theta = \sqrt{3}$

c) $\cos^2 \theta = \frac{3}{4}$

d) $\tan(2\theta) = 1$

e) $\sin\left(\frac{\theta}{3}\right) = 0$

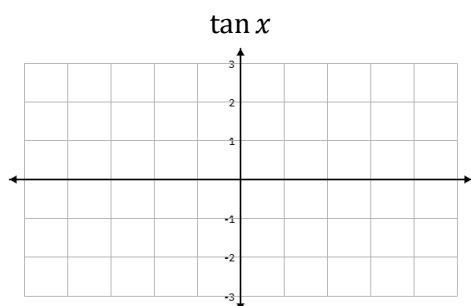
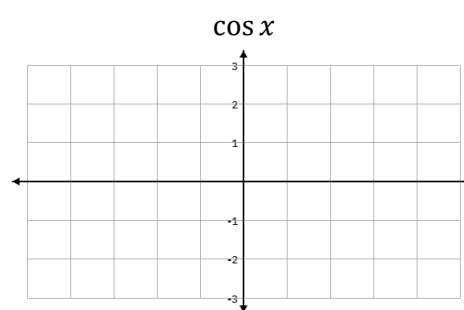
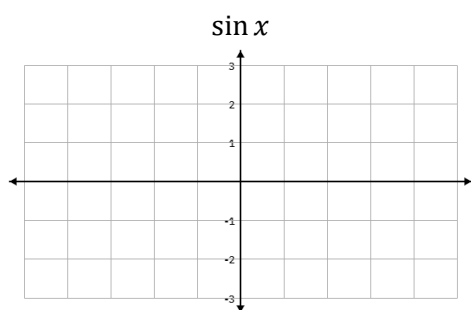
Give the general solution for θ satisfying each equation.

f) $2\sin(3\theta) = \sqrt{3}$

g) $\cos\left(\frac{\theta}{2}\right) = 1$

h) $\tan(6\theta) = -1$

8. On each grid provided, draw the graph of $\sin x$, $\cos x$, and $\tan x$. Label each horizontal gridline to be $\frac{\pi}{2}$ units apart. Remember to draw the asymptotes for the tangent function.



Part II – The Harder Stuff

Wanting to make an A or B on the exam? You'll need to be able to do these!

- Evaluating Trig Functions – Time yourself for 5 minutes! (Below 10 minutes is GREAT).

$\sin\left(\frac{2\pi}{3}\right) =$	$\sin\left(\frac{7\pi}{6}\right) =$	$\cos\left(\frac{\pi}{3}\right) =$	$\sin\left(-\frac{\pi}{2}\right) =$	$\tan\left(\frac{3\pi}{4}\right) =$
$\tan\left(\frac{11\pi}{6}\right) =$	$\cos(3\pi) =$	$\sin\left(\frac{\pi}{4}\right) =$	$\cos\left(\frac{3\pi}{2}\right) =$	$\tan\left(\frac{\pi}{3}\right) =$
$\sin\left(-\frac{11\pi}{6}\right) =$	$\tan\left(\frac{5\pi}{6}\right) =$	$\cos\left(\frac{5\pi}{4}\right) =$	$\sin(\pi) =$	$\cos\left(\frac{8\pi}{3}\right) =$
$\sec\left(\frac{\pi}{3}\right) =$	$\csc\left(-\frac{\pi}{6}\right) =$	$\cot\left(\frac{7\pi}{4}\right) =$	$\sin\left(\frac{23\pi}{6}\right) =$	$\cos\left(-\frac{10\pi}{3}\right) =$

- Simplifying Trig Expressions

$\frac{\cos^2 \theta}{1 - \cos^2 \theta}$	$\frac{1 + \cot \theta}{\csc \theta}$	$\cos \theta \csc \theta (\sec^2 \theta - 1)$
$\frac{\sin^2 \theta \cot^2 \theta}{1 - \sin^2 \theta}$	$\frac{1}{\sec \theta - \tan \theta} - \frac{1}{\sec \theta + \tan \theta}$	$\sin \theta \tan \theta - \csc \theta \tan \theta$
$\frac{\sin^2 \theta - \tan^2 \theta}{\tan^2 \theta \cdot \sin^2 \theta}$	$\frac{(\sin \theta + \tan \theta)^2 + \cos^2 \theta - \sec^2 \theta}{\tan \theta}$	$\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} + 1$
$\frac{\tan \theta - \tan \theta \sin^2 \theta}{2 \sin \theta \cos \theta}$	$\frac{\sin \theta}{1 - \cos \theta} + \frac{1 - \cos \theta}{\sin \theta}$	$\frac{\sin^3 \theta + \cos^3 \theta}{1 - \sin \theta \cdot \cos \theta}$

- Solving Trig Equations – Solve for $0 \leq \theta < 2\pi$.

$\sec \theta \cdot \sec \theta - 2 \sin \theta = 0$	$\sqrt{2} \sin \theta = 2 \cos \theta \cdot \sin \theta$	$\cos^2(2x) = \frac{3}{4}$
$\sin^3 \theta - \frac{\sin \theta}{4} = 0$	$2 \tan \theta \cdot \cos(2\theta) = 2 \cos(2\theta)$	$2 \sin x \cdot \cos x - \sin x + 2 \cos x - 1 = 0$